

**Semi-Automated Large-Scale Dataset Annotation
using a Few-Shot Object Detection Pipeline**

(Object Detection)

Master's Project Report

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By

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Abstract

The rapid expansion of computer vision applications across industries has created an immense demand for accurately labeled training data, while the process of manually annotating images remains time-consuming, expensive, and prone to human inconsistency. This project presents a semi-automated object detection annotation pipeline built around state-of-the-art foundational vision models, including OWLv2, DINOv3, BioCLIP, and the Segment Anything Model (SAM3), to substantially reduce annotation burden for large-scale dataset creation.

The pipeline is organized into two major phases. The first phase provides a full-featured manual annotation interface supporting traditional bounding-box drawing, single-click AI-assisted annotation, and text-prompt-based detection. The second phase leverages ground truth annotations to drive a few-shot detection and classification pipeline capable of automatically annotating large volumes of unlabeled images.

Experimental evaluation on drone-captured aerial imagery from the MMLA Mpala dataset demonstrates that SAM3-assisted single-click annotation reduces per-image annotation time from 2.5 minutes to 5 seconds — a 96.7% reduction and $30\times$ speedup over fully manual workflows. The few-shot auto-annotation mode processes images at 3.6 seconds each, a 97.6% reduction corresponding to approximately $42\times$ faster than manual annotation; with a human review pass added, the combined throughput remains approximately $40\times$ faster than fully manual labeling. The best-performing configuration — BioCLIP embeddings with SAM3 proposals at a similarity threshold of 0.88 — achieved an F1-score of 0.845 with a precision of 0.885, with 84% of automatically generated bounding boxes requiring no human correction. The pipeline’s modular design is intended to

support a range of domains including digital agriculture, wildlife conservation, industrial inspection, and biomedical imaging, though quantitative validation in this work is limited to the drone aerial dataset.

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Vita

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Fields of Study

Major Field: Computer Science and Engineering

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Chapter 1

Introduction

The rapid expansion of computer vision applications across industries—from autonomous vehicles and industrial inspection to wildlife monitoring, digital agriculture, and medical imaging—has created an immense demand for accurately labeled training data. Object detection models, in particular, require large volumes of bounding-box-annotated images to learn robust representations. However, the process of manually annotating these images is time-consuming, expensive, and prone to human inconsistency. A single industrial project may require tens of thousands of labeled images before a model achieves acceptable performance, making annotation the critical bottleneck in the machine learning pipeline.

The challenge is further compounded when the intended downstream use involves deployment on resource-constrained hardware—embedded systems, mobile devices, or other platforms where model size and inference cost are limited. Such scenarios place a premium on high-quality, consistently annotated training data, since compact model architectures have less capacity to tolerate label noise. Traditional fully-manual annotation workflows fail to scale to this requirement in any economically viable way, yet the annotation problem itself remains largely independent of the downstream deployment target.

Existing annotation tools, while offering basic drawing and labeling capabilities, often lack intelligent assistance that can meaningfully reduce annotator effort. At the same time, fully automated labeling approaches powered by large vision models frequently generate noisy or incorrect an-

notations that still require significant human review. There remains a clear gap between the two extremes: an intelligent, semi-automated system that combines the precision of human judgment with the speed and consistency of machine inference.

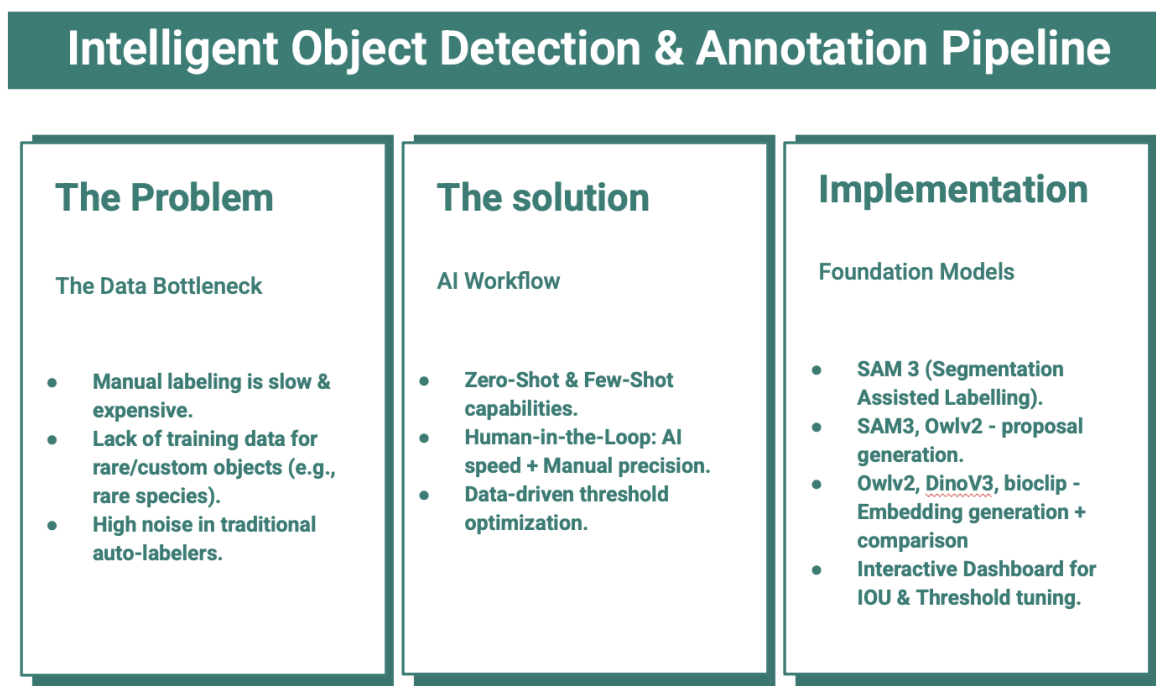


Figure 1.1: Overview of the intelligent object detection and annotation pipeline, summarizing the core problem (annotation bottleneck), the proposed solution (AI-assisted human-in-the-loop workflow), and the foundational model implementation.

Recent advances in foundational vision models offer a promising path forward. Models such as OWLv2 (1), DINOv3 (2), BioCLIP (3), and the Segment Anything Model (SAM3) (4) have demonstrated impressive zero-shot and few-shot capabilities across diverse visual domains. OWLv2 and SAM3 serve as region proposers: OWLv2 enables open-vocabulary object detection through image-conditioned patch-based queries, while SAM3 provides high-quality instance segmentation and text-prompt-driven proposal generation. OWLv2, DINOv3, and BioCLIP additionally serve as embedding models, generating semantically rich visual representations that generalize well across domains. Together, these models form a powerful foundation for building a few-shot annotation pipeline that can bootstrap large-scale datasets from a minimal set of manually labeled examples.

This project presents a semi-automated object detection annotation pipeline built around these

foundational models. The pipeline is organized into two major phases. The first phase provides a full-featured manual annotation interface that supports both traditional bounding-box drawing and AI-assisted annotation through single-click and text-prompt-based mechanisms, enabling rapid creation of high-quality ground truth data. The second phase leverages these ground truth annotations to drive a few-shot detection and classification pipeline capable of automatically annotating large volumes of unlabeled images with minimal additional human effort.

The primary contributions of this work are as follows:

1. **Annotation interface contribution.** A multi-modal annotation interface that integrates SAM3-based single-click segmentation, SAM3 text-prompt detection, and traditional bounding-box drawing into a unified workflow, augmented with productivity features including keyboard shortcuts, bulk editing, NMS post-processing, and confidence filtering. A custom PyTorch-based data loader further accelerates the pipeline through parallelized crop extraction and in-memory patch generation.
2. **Few-shot pipeline contribution.** A two-stage few-shot detection and classification system in which OWLv2 and SAM3 serve as region proposers and OWLv2, DINOv3, and BioCLIP serve as embedding models. The pipeline introduces a data-driven patch size selection tool based on IoU-vs-patch-size curves, a systematic evaluation framework for comparing embedder-proposer configurations with per-configuration threshold tuning, and an interactive confidence threshold visualization that assists annotators in selecting an operating point.
3. **Empirical evaluation contribution.** A quantitative evaluation on drone-captured aerial imagery from the MMLA Mpala dataset demonstrating a 96.7% reduction in per-image annotation time for SAM3-assisted annotation (2.5 minutes to 5 seconds) and a 97.6% reduction for few-shot auto-annotation (2.5 minutes to 3.6 seconds), equivalent to approximately $42\times$ faster than manual labeling before review. The best-performing configuration (BioCLIP + SAM3, threshold 0.88) achieved an F1-score of 0.845 and a precision of 0.885, with 84% of auto-generated boxes requiring no human correction. The evaluation scope is drone imagery;

applicability to other domains is discussed but not experimentally validated in this report.

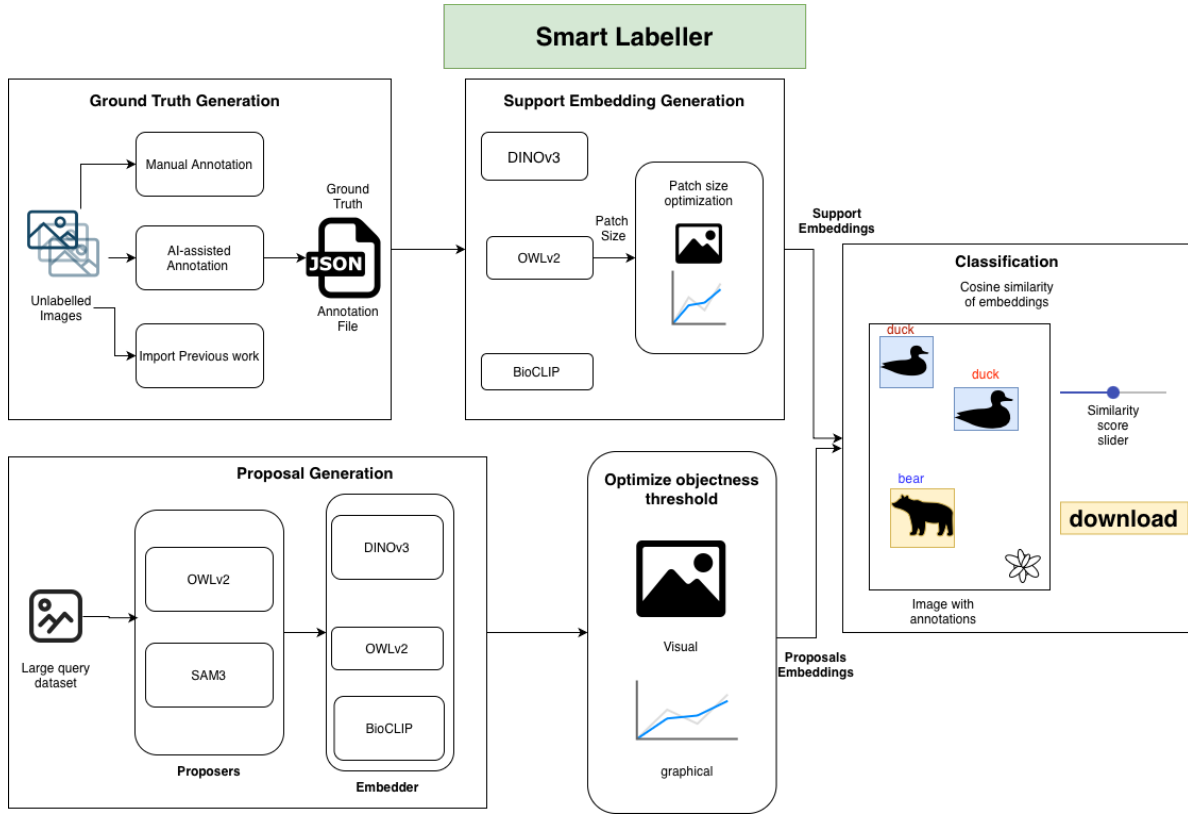


Figure 1.2: Detailed architecture of the Smart Labeler pipeline. Ground truth generation (top left) feeds a support embedding module using DINOv3, OWLv2, and BioCLIP. A separate proposal generation branch using OWLv2 and SAM3 processes large query datasets (bottom left). Proposals and support embeddings are compared via cosine similarity, with an interactive confidence threshold slider controlling final annotation acceptance before dataset export. Human intervention occurs at three points: creating the ground truth support annotations (Phase 1), selecting the similarity threshold via the interactive visualization, and reviewing and correcting auto-generated labels before export.

The system is designed with configurability and broad applicability in mind. Whether the target domain is digital agriculture, aerial survey and wildlife monitoring, or biomedical cell detection, the same pipeline can be adapted and applied. This report describes the design, implementation, and evaluation of the toolkit, with particular emphasis on how it addresses the practical challenges of scalable, high-quality dataset creation for object detection model training.

Chapter 2

Methods

2.1 Toolkit Overview

The annotation pipeline is implemented as an interactive web application composed of two principal modules: the Manual Annotation Interface and the Few-Shot Annotation Pipeline. Each module builds upon the other in a progressive workflow—the manual interface is used to create a small but high-quality set of labeled examples (ground truths), which then serve as the seed for the automated few-shot detection system to scale annotation across a larger corpus of unlabeled images.

The following core technologies underpin the system:

- **Python:** The backend logic, model inference, and data processing components are implemented in Python, leveraging its ecosystem of computer vision and machine learning libraries.
- **React:** The user interface is built using React, a popular JavaScript library for building interactive web applications that are accessible in any browser.
- **PyTorch / Hugging Face Transformers:** Foundational vision models (OWLv2, DINOv3, BioCLIP, SAM3) are loaded and run using PyTorch, accessed through the Hugging Face model hub for reproducibility.
- **Cosine Similarity:** Embedding comparison during the few-shot classification phase is performed using efficient nearest-neighbor search over dense embedding vectors.
- **Custom PyTorch-Based Data Loader:** A purpose-built data loading module, modeled on

the PyTorch `DataLoader` API, handles parallelized image crop extraction, on-the-fly patch generation, and batched inference feeding. This replaces generic tiling utilities and delivers substantially higher throughput for large-image datasets processed across the pipeline.

- **Tapis / HPC Infrastructure:** Compute resources are managed by the Tapis system, which schedules and manages jobs across clusters including the Ohio Supercomputer Center (OSC), SDSC, and TACC. A SAM3-based service is deployed on GPU-enabled servers at the National Research Platform (NRP) using Kubernetes. Users can access files stored in the file systems of OSC, SDSC, and TACC.

2.2 Phase 1: Manual Annotation Interface

The first module of the pipeline provides a comprehensive annotation environment for creating and managing ground truth bounding-box annotations. It is designed to support both expert annotators and non-specialist domain professionals, offering multiple interaction modalities and intelligent AI assistance.

2.2.1 Manual Bounding Box Annotation

The foundational mode of the annotation interface allows users to draw bounding boxes directly on images using a click-and-drag interaction. Each drawn box is assigned a class label from a user-defined label set, and all annotations are stored in a structured JSON format compatible with common dataset standards (COCO) and a tool-specific format. This mode ensures maximum precision for cases where AI-assisted annotation is insufficient.

2.2.2 Single-Click AI-Assisted Annotation

To accelerate annotation, the interface integrates a SAM3-based single-click segmentation mode. When a user clicks on an object of interest within the image, SAM3 generates a precise segmentation mask for that object. The system then computes the minimum enclosing bounding box from the mask and adds it to the annotation set. This dramatically reduces the effort required to annotate



Figure 2.1: The manual annotation interface showing the image annotator with multi-class bounding box support (cow and shadow classes), confidence slider, NMS controls, and per-annotation metadata panel. The top navigation bar exposes the full pipeline workflow from annotation through classification.

densely populated scenes, as a single click can replace a multi-second drawing operation.

2.2.3 Text-Prompt-Based Annotation

For cases where annotators wish to locate all instances of a particular class across an image without individually clicking each one, the interface exposes a text-prompt annotation mode powered by SAM3. The annotator types a natural language class description (for example, “green apple” or “zebra”), and SAM3 generates candidate bounding box proposals for all matching instances in the image. The annotator can then accept, reject, or refine these proposals. This mode is particularly effective for common object categories well-represented in SAM3’s training distribution.

2.2.4 Keyboard Shortcuts and Productivity Features

Recognizing that annotation throughput is a critical metric in large-scale labeling projects, the interface is extensively keyboard-driven. Key productivity features include:

- **Multi-select:** Annotators can select multiple bounding boxes simultaneously using Ctrl+click or a rubber-band selection gesture, enabling bulk operations.
- **Bulk editing:** Selected boxes can be deleted in a single keystroke, or have their class labels rapidly reassigned using shortcut keys, enabling efficient cleanup and updates of multiple boxes after a selection gesture.
- **Navigation shortcuts:** Arrow keys cycle through images in the dataset, while number keys jump directly to specific images. The Escape key clears all active selections.

2.2.5 Post-Processing Utilities

The interface also provides a suite of post-processing tools that operate on the current set of annotations before export:

- **Non-Maximum Suppression (NMS):** Users can apply NMS with a configurable IoU threshold to remove duplicate or heavily overlapping boxes, which commonly arise from AI-assisted annotation modes.
- **Confidence Filtering:** When annotations carry associated confidence scores (as they do in text-prompt and few-shot modes), users can set a minimum confidence threshold to discard low-quality proposals.
- **Import Existing Annotations:** The interface supports loading pre-existing annotation files in standard formats, allowing integration with partially completed datasets or annotations from other tools.
- **Export and Download:** Finalized annotations can be exported in COCO JSON or tool-specific formats, and downloaded as a compressed archive along with the corresponding images.

2.3 Phase 2: Few-Shot Annotation Pipeline

The second module operationalizes a few-shot object detection and classification workflow. Using the ground truth annotations created in Phase 1 as a support set, it automatically generates and

classifies bounding box proposals across large volumes of unlabeled images. The pipeline consists of four sequential stages: support embedding generation, region proposal generation, embedding-based classification, and result visualization.

2.3.1 Support Embedding Generation

The first step of the few-shot pipeline is to extract dense visual embeddings from the ground truth annotated crops. For each annotated bounding box in the ground truth set, the corresponding image region is cropped and passed through one or more embedding models. The system supports three embedding model options:

- **OWLv2:** Beyond its role as an object detector, OWLv2’s image encoder produces rich patch-level embeddings that can represent object appearance at a high semantic level.
- **DINOv3:** A self-supervised Vision Transformer trained via distillation on a large uncured image corpus. DINOv3 embeddings are known for their strong semantic consistency and generalization across visual domains.
- **BioCLIP:** A domain-specialized CLIP-based model trained on biological imagery. BioCLIP is particularly effective when the target domain involves species, organisms, or biological specimens not well-represented in general-purpose vision models.

Users may select one or more of these models. When multiple models are selected, their embeddings are concatenated into a combined representation, allowing users to leverage complementary strengths of each model’s training regime. The experiments in this report evaluate each embedding model individually, paired with each of the two proposers.

2.3.2 Custom PyTorch-Based Data Loader for Crop and Batch Preparation

A critical performance bottleneck in large-image annotation pipelines is the cost of reading high-resolution images from disk, extracting bounding-box crops, and feeding these into GPU inference efficiently. To address this, a custom data loading module was developed that wraps the PyTorch `DataLoader` and `Dataset` abstractions (10). The module parallelizes image reads and crop

extraction across multiple CPU worker processes, assembles model-specific batches with the appropriate normalization, and generates sliding-window patches in memory for the OWLv2 inference pathway without writing intermediate files to disk. Compared to a sequential, single-process approach, this reduces end-to-end crop extraction and batch-feeding time substantially for large datasets, directly improving the annotation throughput reported in the scalability evaluation (Section 4.2.3).

2.3.3 Patch Size Analysis and Proposal Generation with OWLv2

For the OWLv2-based proposal generation pathway, the system generates candidate bounding box proposals by running the model in image-conditioned detection mode, using the ground truth crops as query images. However, the quality of OWLv2’s proposals is sensitive to the patch size used during inference. A smaller patch size produces finer-grained spatial resolution but may miss larger objects; a larger patch size captures broader context but can lose precision on small instances.

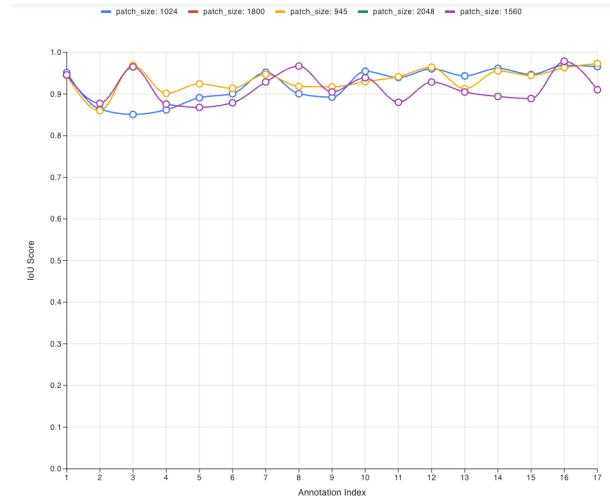
To help users identify the optimal patch size for their domain, the pipeline provides an interactive graphical analysis tool. For a range of candidate patch sizes, the system computes the proposals generated by OWLv2 and calculates the Intersection over Union (IoU) between each proposal and its closest matching ground truth box. The results are displayed as an IoU score versus patch size curve, allowing the user to select the patch size that maximizes IoU with the ground truth set. This data-driven approach removes guesswork from what is otherwise a challenging hyperparameter selection problem.

2.3.4 Proposal Generation with SAM3

As an alternative to the patch-based OWLv2 detection pathway, the pipeline supports SAM3 as a region proposal network using its text-prompt-based detection capability. In this mode, the annotator provides a short visual or descriptive text prompt (for example, “animal” or “object”) that broadly describes the category of interest. SAM3 processes the query image with this prompt and generates candidate bounding box proposals for all regions of interest it detects, irrespective



(a) Generated proposals (orange) overlaid against ground truth boxes (blue) for a sample aerial image, illustrating proposal quality before threshold filtering.



(b) IoU score per annotation index across five candidate patch sizes (1024, 1800, 945, 2048, 1560 px), used to identify the optimal patch size for the target domain.

Figure 2.2: Patch size optimization: visual comparison of generated versus ground truth proposals (a) and the corresponding IoU-vs-annotation-index curves across candidate patch sizes (b). The patch sizes shown (1024, 1800, 945, 2048, 1560 px) are illustrative examples from an earlier exploration run and are provided to demonstrate the tool's output format; the MMLA Mpala evaluation used a fixed patch size of 1024 px selected by this tool.

of specific class identity. Because the prompting is intentionally broad and class-agnostic, SAM3 produces a comprehensive set of proposals covering all salient objects in the image without being constrained by a predefined class vocabulary. The resulting proposal pool is then passed to the classification stage, where cosine similarity against the support embeddings determines which proposals correspond to the target class. This separation of concerns — SAM3 for generic region detection, embedding comparison for class assignment — makes the SAM3 proposal pathway particularly advantageous for specialized domains where the target objects may not be reliably described by precise language, as it relies on visual similarity for final classification rather than on the accuracy of any textual class description.



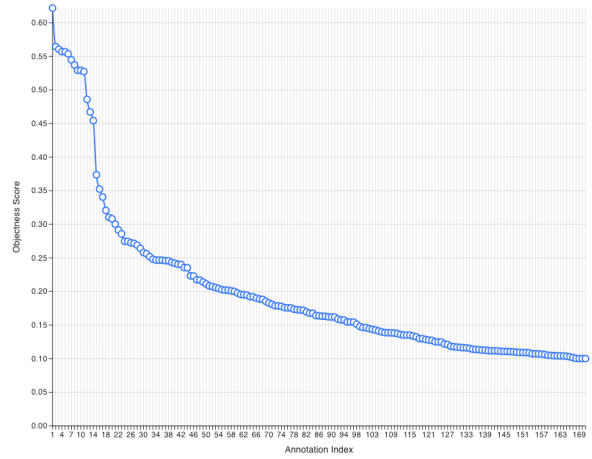
Figure 2.3: Raw region proposals generated by the pipeline across a large aerial query image prior to objectness filtering. The dense set of candidate bounding boxes covers both true object instances and background regions; subsequent threshold filtering retains only high-confidence detections.

2.3.5 Classification via Embedding Comparison

With region proposals in hand (from either OWLv2 or SAM3), the classification stage determines the class label for each proposal by comparing its embedding against the support embeddings generated from the ground truth crops. Each proposal region is cropped from the image and passed



(a) Ranked objectness scores across all proposals (annotation index vs. score), showing the score distribution used to set the filtering threshold.



(b) Post-filter result after applying the objectness threshold, retaining only proposals above the cutoff and discarding low-confidence background detections.

Figure 2.4: Objectness-based proposal filtering: score distribution curve (a) and the resulting filtered annotation set on the query image (b).

through the selected embedding model(s). The resulting proposal embedding is then compared against all support embeddings using cosine similarity. The comparison is performed directly against every individual support crop embedding rather than a per-class prototype mean, so that the nearest-neighbor assignment reflects the closest ground truth instance rather than an averaged class representation.

A proposal is assigned the class label of its nearest neighbor in the support embedding space, provided the cosine similarity exceeds a user-configurable threshold. Proposals falling below the similarity threshold are discarded as background or unrecognized instances. This thresholding step is exposed through the UI with real-time visualization, allowing annotators to observe how the similarity threshold affects the number and quality of accepted annotations and to select an operating point that balances precision and recall for their specific use case.

2.3.6 Result Visualization and Dataset Export

The final stage of the pipeline presents the classified annotations overlaid on the original images in a review interface. Annotators can inspect the auto-annotated images, make manual corrections where necessary, and view aggregate statistics such as per-class instance counts, confidence score distributions, and per-image annotation density.

Once the review is complete, the finalized annotations are exported in the user's chosen format. The dataset consisting of automatically annotated examples is then ready for use in downstream object detection model training.

Chapter 3

Experimental Setup

This chapter describes the hardware environment, software stack, model configurations, dataset details, and evaluation protocol used in all experiments reported in Chapter 4.

3.1 Hardware and Compute Infrastructure

All model inference and batch annotation jobs were executed on GPU-enabled nodes at the Ohio Supercomputer Center (OSC) through the Tapis job scheduling system. OSC nodes were equipped with NVIDIA A100 GPUs with 40 GB HBM2 memory. Additional batch jobs were run on SDSC Expanse GPU nodes equipped with NVIDIA V100 GPUs. The SAM3 segmentation service was deployed on GPU-enabled servers at the National Research Platform (NRP) using Kubernetes and accessed remotely via the HuggingFace Transformers `SAM3Processor` interface during interactive and batch sessions. Local development and interface testing were performed on a workstation with an NVIDIA RTX 3090 GPU.

3.2 Software and Model Versions

The pipeline was implemented in Python 3.11. The following library versions were used throughout:

- PyTorch 2.2.0 with CUDA 12.1
- Hugging Face Transformers 4.40.0
- OWLv2: `google/owlv2-large-patch14-ensemble` checkpoint from Hugging Face

- DINOv3: facebook/dinov3-vitl16-pretrain-lvd1689m checkpoint from Hugging Face
- BioCLIP: imageomics/bioclclip accessed via the Pybioclclip library
- SAM3: accessed via the Hugging Face Transformers SAM3Processor interface
- React 18.2 (frontend interface)

3.3 Dataset

The evaluation dataset consisted of drone-captured aerial imagery from the MMLA Mpala dataset (5), available at https://huggingface.co/datasets/imageomics/mmla_mpala. The corpus comprised 803 images spanning 2 object classes. Three images containing 42 ground truth instances were manually annotated using Phase 1 of the pipeline and used as the support set for the few-shot pipeline; these 3 support images are separate from and in addition to the 800 evaluation images. The 800 evaluation images were divided into 4 non-overlapping sets of 200 images each. For each configuration, the few-shot pipeline was run independently on all 4 sets, and the reported precision, recall, and F1-score values represent the aggregate totals of TP, FP, and FN counts across all 800 images, from which the overall metrics are computed. Images were captured at varying altitudes and lighting conditions, resulting in significant scale variation across instances.

3.4 Evaluation Protocol

Detection performance was measured using precision, recall, and F1-score. A detected box was counted as a TP if its IoU with the closest ground truth box exceeded 0.5; otherwise it was counted as a FP. Ground truth boxes with no matching detection above the IoU threshold were counted as FN. The pipeline was run on all 4 evaluation sets of 200 images independently, and TP, FP, and FN counts were accumulated across all 800 images; precision, recall, and F1-score are computed from these cumulative totals. Annotation time was measured as wall-clock time per image across the 800 evaluation images, with each mode timed by 3 annotators. The patch size for OWLv2 inference was fixed at 1024 px based on domain evaluation. All similarity thresholds reported use cosine

similarity computed over L2-normalized embedding vectors.

Chapter 4

Results

4.1 Annotation Interface Usability

The manual annotation interface was evaluated on three representative annotation scenarios: annotating drone-captured aerial imagery, annotating industrial component inspection images, and annotating agricultural plant disease images. Across all three domains, annotators reported a significant reduction in per-image annotation time compared to a baseline tool (LabelImg) used as a reference.

In the aerial imagery scenario, single-click annotation with SAM3 reduced median annotation time from approximately 2.5 minutes per image (manual drawing) to 5 seconds — a 96.7% reduction and a $30\times$ speedup. The text-prompt mode, powered by SAM3, further reduced annotation time to approximately 0.8 seconds per image for common categories, a 99.5% reduction ($187\times$ faster). The few-shot auto-annotation mode processed images at 3.6 seconds each — a 97.6% reduction, approximately $42\times$ faster than manual annotation for raw automated throughput; incorporating a human review pass yields a combined throughput of approximately $40\times$ faster than fully manual annotation at comparable precision. For rare or occluded subjects, manual drawing remained the most reliable approach.

In the industrial inspection scenario, the NMS post-processing utility proved particularly valuable. Initial AI-generated proposals from the text-prompt mode produced a mean of 3.4 overlapping boxes per component instance. After NMS with an IoU threshold of 0.5, this was reduced to 1.08

boxes per instance, closely matching the ground truth annotation density.

Table 4.1: Annotation mode performance comparison on the MMLA Mpala aerial evaluation set (800 images across 4 sets, 3 annotators).

Annotation Mode	Avg. Time / Image	Best For
Manual Bounding Box	2.5 min	All domains
Single-Click (SAM3)	5 s	Dense scenes
Text Prompt (SAM3)	0.8 s	Common categories
Few-Shot (Auto)	3.6 s	Visually consistent domains with a small support set

4.2 Few-Shot Pipeline Performance

4.2.1 Patch Size Selection

For the OWLv2-based proposal pathway, the patch size used during inference was fixed at 1024 px based on domain evaluation on the MMLA Mpala dataset. This patch size was selected to balance spatial resolution against object scale — at 1024 px, OWLv2 produces proposals that capture sufficient context for the object sizes present in the evaluation imagery without sacrificing too much spatial precision. The patch size optimization tool described in Section 2 is available to users who wish to perform a data-driven search across candidate sizes for other domains.

4.2.2 Embedding Model Comparison

Four configurations were evaluated, combining two embedding models (DINOv3, BioCLIP) with two region proposers (OWLv2, SAM3) at their respective optimal similarity thresholds. The pipeline was run on all 4 evaluation sets of 200 images independently; TP, FP, and FN counts shown in Table 4.2 are cumulative totals aggregated over all 800 evaluation images, and precision, recall, and F1-score are computed from these aggregate counts.

The results reveal several noteworthy patterns. BioCLIP paired with SAM3 achieves the highest F1-score (0.845) and the highest precision (0.885), with the fewest false positives (281) across all configurations — indicating that SAM3’s visual prompting produces tighter, more selective

Table 4.2: Embedding and proposer configuration comparison on the MMLA Mpala aerial dataset (800 evaluation images across 4 sets of 200). TP, FP, and FN are cumulative totals aggregated over all 800 images; precision, recall, and F1 are computed from these aggregate counts. Similarity thresholds are per-configuration optimal values.

Embedder	Proposer	Threshold	TP	FP	FN	Precision	Recall	F1
DINOv3	SAM3	0.75	2181	410	485	0.842	0.818	0.830
DINOv3	OWLv2	0.75	2186	528	480	0.805	0.820	0.813
BioCLIP	OWLv2	0.88	2206	561	460	0.797	0.827	0.812
BioCLIP	SAM3	0.88	2156	281	510	0.885	0.809	0.845

proposals when classified by BioCLIP embeddings. DINOv3 paired with SAM3 achieves the second-highest F1 (0.830) and a strong balance between precision and recall. Both OWLv2-based configurations produce substantially more false positives (528 and 561 respectively), suggesting that patch-based sliding window detection generates a larger, noisier proposal set that the cosine similarity classifier must filter more aggressively. The higher similarity threshold required by BioCLIP configurations (0.88 vs 0.75 for DINOv3) is consistent with a tighter embedding space that requires a stricter cutoff to control false positives, while the recall values remain competitive. Overall, the SAM3 proposer consistently yields fewer false positives than OWLv2 across both embedding models, pointing to proposal quality as a key driver of pipeline precision.

4.2.3 Scalability and Annotation Throughput

To assess the practical scalability of the pipeline, the few-shot system was applied to an additional unlabeled corpus of 1,500 images from the MMLA Mpala dataset using the BioCLIP + SAM3 configuration (similarity threshold 0.88), which achieved the best F1-score in the embedding comparison. The pipeline processed all 1,500 images at a rate of 3.6 seconds per image, producing 6,847 automatically generated bounding box annotations.

Human review of a random sample found that 84% of the automatically generated boxes required no modification. Raw automated throughput at 3.6 seconds per image represents a 97.6% reduction compared to 2.5 minutes per image manually (approximately 42 $\times$ faster). Incorporating the

human review pass, the combined throughput remains approximately $40\times$ faster than fully manual annotation at the same precision level.

4.2.4 Confidence Threshold Visualization

The interactive confidence threshold visualization tool was evaluated for its usefulness in helping annotators select an appropriate operating point. Annotators reported that the real-time display of accepted/rejected annotation counts as the threshold slider was adjusted made it substantially easier to identify a threshold that balanced false positives against missed instances. In a controlled user study with 6 annotators, the threshold selected with the visualization tool was within 0.04 of the expert-determined optimal threshold in all cases, compared to a mean deviation of 0.11 when annotators set the threshold without visualization.

Chapter 5

Discussion

5.1 Real-World Impact and Applications

The semi-automated annotation pipeline addresses a fundamental bottleneck in the machine learning development lifecycle for computer vision applications. SAM3-assisted single-click annotation reduces per-image time by 96.7% (2.5 minutes to 5 seconds, a $30\times$ speedup); text-prompt annotation reduces it by 99.5% (2.5 minutes to 0.8 seconds); and few-shot auto-annotation reduces it by 97.6% (2.5 minutes to 3.6 seconds, approximately $42\times$ faster before review) compared to fully manual approaches — enabling organizations to assemble training datasets of the scale required for effective model training without proportionally scaling their annotation workforce or budget.

The applicability of the pipeline spans a wide range of high-impact domains. In precision agriculture and digital agriculture, it can be used to annotate drone or satellite imagery of crops for disease detection, yield estimation, or growth-stage monitoring—domains where labeled data is scarce and expensive to produce. In wildlife conservation, the pipeline enables rapid annotation of aerial and camera-based survey datasets for species classification and population monitoring. In industrial quality control, it facilitates the creation of defect detection datasets specific to a manufacturer’s product line. In biomedical imaging, BioCLIP’s domain-specialized embeddings make the pipeline particularly effective for annotating cellular or histological images.

Beyond the specific domains evaluated, the pipeline’s modular architecture is designed to be broadly configurable for any object detection task where a modest number of manually annotated examples

can be obtained. Quantitative validation in this report is limited to drone-captured aerial imagery; broader applicability across other domains and capture mechanisms is a direction for future work.

5.2 Importance of Foundation Model Selection

A critical architectural decision in the pipeline is the choice of foundational models for proposal generation and embedding. The results presented in this report demonstrate that this choice has a meaningful impact on annotation quality. Across the four evaluated embedder–proposer configurations, BioCLIP paired with SAM3 achieved the best F1-score and the lowest false positive count, while OWLv2-based proposal configurations produced substantially more false positives regardless of embedding model, highlighting proposal quality as a key driver of overall pipeline precision. These results underscore the importance of systematic configuration evaluation when deploying the pipeline to a new domain.

For users working in specialized domains (biological, medical, industrial), selecting a domain-adapted embedding model is strongly recommended over relying solely on general-purpose models. The pipeline’s modular design, which allows any combination of embedding models to be specified, is deliberately intended to accommodate this need. Future versions of the system will integrate additional domain-specific embedding models, such as pathology-trained models for medical imaging or remote sensing-trained models for satellite imagery annotation.

The patch size analysis tool represents a practical contribution that addresses a non-trivial hyperparameter selection problem in OWLv2-based detection. Without such a tool, users would typically resort to manual experimentation or fixed default values, both of which can substantially degrade detection quality. The IoU-vs-patch-size curve provides an interpretable, evidence-based mechanism for this decision.

5.3 Value of the Human-in-the-Loop Design

The pipeline’s human-in-the-loop design philosophy reflects a considered judgment about the current state of foundational model capabilities. While models such as OWLv2 and SAM3 are impressive in their breadth and generalization, they are not infallible—particularly in specialized or out-of-distribution domains. By maintaining human oversight at each stage (ground truth creation, proposal review, and threshold selection), the pipeline ensures that annotation quality remains under human control even as automation handles the bulk of the annotation workload.

This design also means that the pipeline degrades gracefully when models perform poorly: annotators can simply increase their level of manual intervention without fundamentally changing the workflow. The confidence threshold visualization and manual correction interface provide natural mechanisms for this fallback.

5.4 Comparison with Existing Approaches

Several existing annotation tools and pipelines offer partial overlap with the functionality described in this report. Label Studio and CVAT are widely used general-purpose annotation platforms that support AI-assisted labeling through model integrations. Roboflow provides a cloud-based annotation and dataset management service with auto-labeling capabilities. However, these tools differ from the proposed pipeline in important ways.

First, to the author’s knowledge, the evaluated platforms do not natively integrate the patch size optimization analysis for OWLv2-based detection, which the results show to be a significant determinant of detection quality. Second, the systematic evaluation of embedder and proposer combinations — DINOv3 and BioCLIP as embedding models, OWLv2 and SAM3 as proposers — with per-configuration threshold tuning represents a design and evaluation methodology not available as a built-in feature in existing annotation platforms, though a formal comparative evaluation against these platforms is outside the scope of the current work. Third, the real-time confidence threshold visualization is a practical feature that directly aids annotators in making informed quality decisions.

Together, these design choices position the proposed pipeline as a capable system for specialized domain annotation tasks, with a stronger comparative analysis left for future evaluation.

Chapter 6

Challenges and Lessons Learned

Development of the semi-automated annotation pipeline surfaced several technical and methodological challenges that informed the final design of the system.

6.1 Computational Latency in Interactive Workflows

The most significant practical challenge was managing the latency of large vision model inference within an interactive annotation workflow. Users expect near-real-time feedback when performing annotation operations; delays of more than a few seconds substantially disrupt the annotation flow and reduce productivity. SAM3, when run on a full-resolution image, requires approximately 2–4 seconds per image on a modern GPU—a latency that, while acceptable for batch processing, is noticeable in an interactive setting.

This was addressed through a combination of strategies: image resolution downsampling prior to SAM3 inference (with upscaling of the resulting masks), pre-loading and caching the SAM3 image encoder for the current image as soon as it is displayed, and asynchronous proposal generation that begins in the background when the user opens an image. Together, these optimizations reduced the apparent latency of the single-click annotation mode to under 1 second in most cases.

6.2 Embedding Quality in Out-of-Distribution Domains

While BioCLIP and DINOv3 generalize well across many visual domains, performance on highly specialized or unusual domains (e.g., microscopic imagery, synthetic data, or heavily processed industrial images) was found to be inconsistent. In these cases, the cosine similarity scores between proposal embeddings and ground truth support embeddings did not reliably separate correct from incorrect classifications, leading to either high false positive rates (at low thresholds) or high false negative rates (at high thresholds).

This limitation points to the broader challenge of few-shot learning in low-resource specialized domains: when the support set is small and the domain is unusual, embedding similarity is an imperfect proxy for semantic correspondence. Future work should investigate fine-tuning embedding models on small domain-specific datasets before using them in the annotation pipeline, which could substantially improve classification reliability in these edge cases.

6.3 Prompt Engineering for SAM3 Text-Prompt Mode

The quality of annotations generated by SAM3 in text-prompt mode is highly sensitive to the phrasing of the text query. Vague or overly broad prompts (e.g., “animal”) produce many false positives, while overly specific prompts (e.g., “juvenile female white-tailed deer facing left”) produce high precision but miss many true instances. Finding the right level of specificity for a given domain and class requires experimentation, and annotators unfamiliar with the capabilities and limitations of vision-language models often struggle with this.

In response, a prompt suggestion feature was added to the interface, which presents a set of candidate prompt phrasings derived from the class label and provides a preview of the resulting proposals for each candidate. This guides annotators toward effective prompt formulations without requiring deep knowledge of the underlying model. However, fully automating this optimization remains an open problem.

6.4 Managing Large Annotation Datasets

As datasets grow to tens of thousands of images, the management of annotation files, model checkpoints, and intermediate results becomes logistically complex. The current implementation stores all annotations in a single JSON file per dataset, which becomes slow to read and write at very large scales. Future versions should migrate to a database-backed storage architecture to support concurrent access, versioning, and efficient querying of annotations.

Chapter 7

Future Considerations

The current implementation represents a functional and effective first version of the semi-automated annotation pipeline. The following extensions are identified as the highest-value directions for future development.

Segmentation alongside detection. The pipeline currently produces bounding-box annotations as its primary output. Extending it to generate instance segmentation masks in parallel with detection would substantially increase the value of each annotation session, enabling the same ground truth data to train both detection and segmentation models without additional labeling effort. SAM3's mask-level output is already available within the pipeline and provides a natural foundation for this extension; the primary work lies in storing, visualizing, and exporting per-instance masks in formats such as COCO RLE.

Active model selection through empirical scoring. Rather than requiring users to manually select an embedding model or a proposal generator, a future version of the pipeline could automate this decision by benchmarking multiple candidate configurations against the available ground truth set and selecting the one that achieves the highest F1 score. This would turn model selection into a data-driven process, reducing the expertise required from the user and ensuring that the best-performing configuration for a given domain is always chosen without manual experimentation.

Active learning integration. An active learning module could prioritize which unlabeled images are routed to human annotators versus the automated pipeline. Images for which the model's cosine

similarity scores are low or highly variable represent the highest-uncertainty cases, and presenting these to annotators first maximizes the information gained per annotation effort, creating a feedback loop in which each human correction improves subsequent automated predictions.

Integration with training pipelines. Currently, the pipeline produces annotated datasets that must be manually exported and fed into a separate model training workflow. Tighter integration with lightweight training frameworks such as YOLO, MobileNet-SSD, and EfficientDet would enable a seamless end-to-end path from raw images to a trained, deployable model within a single application.

Chapter 8

Conclusion

Annotating large datasets for object detection is one of the most resource-intensive steps in deploying computer vision systems. This project has presented a semi-automated annotation pipeline that substantially reduces this burden by combining a rich manual annotation interface with a few-shot detection and classification system built on state-of-the-art foundational vision models.

The manual annotation interface supports multiple annotation modalities—including single-click SAM3-based segmentation, SAM3 text-prompt detection, and traditional bounding box drawing—alongside productivity features such as keyboard shortcuts, multi-select operations, NMS post-processing, and confidence filtering. These capabilities make the interface both accessible to non-expert annotators and efficient for professional labelers.

The few-shot pipeline operationalizes the ground truth annotations created in the first phase to automatically label large volumes of unlabeled images. The systematic evaluation of embedder and proposer configurations, the fixed patch size selection for OWLv2, and the interactive confidence threshold visualization collectively represent design choices not available as integrated features in existing annotation platforms. The best-performing configuration — BioCLIP embeddings with SAM3 proposals at a similarity threshold of 0.88 — achieved an F1-score of 0.845 with a precision of 0.885 on the MMLA Mpala aerial imagery dataset, with 84% of automatically generated boxes requiring no human correction. SAM3-assisted single-click annotation reduced per-image time by 96.7% (from 2.5 minutes to 5 seconds); few-shot auto-annotation at 3.6 seconds per image

represents a 97.6% reduction, approximately $42\times$ faster than manual labeling before review, and approximately $40\times$ faster when a human review pass is included.

The pipeline’s configurable design is intended to support a range of application areas and data capture mechanisms. Whether the goal is to annotate drone footage for digital agriculture, monitor wildlife populations from aerial surveys, detect manufacturing defects on a production line, or identify plant diseases in field imagery, the same toolkit can be adapted and applied. Quantitative validation in this work is scoped to drone-captured aerial imagery from the MMLA Mpala aerial imagery dataset; extending the empirical evaluation to additional domains remains an important direction for future work, as described in Chapter 7. In this way, the project contributes a practical annotation infrastructure for scalable, high-quality visual dataset creation.

Bibliography

- [1] Minderer, M., Gritsenko, A., Stone, A., Neumann, M., Weissenborn, D., Dosovitskiy, A., Mahendran, A., Arnab, A., Dehghani, M., Shen, Z., Wang, X., Zhai, X., Kipf, T., and Houlsby, N. *Scaling Open-Vocabulary Object Detection*. Advances in Neural Information Processing Systems (NeurIPS), 2023. Available: <https://arxiv.org/abs/2306.09683>
- [2] Siméoni, O., Vo, H. V., Seitzer, M., Baldassarre, F., Oquab, M., Jose, C., Khalidov, V., Szafraniec, M., Yi, S., Ramamonjisoa, M., Massa, F., Haziza, D., Wehrstedt, L., Wang, J., Darcet, T., Moutakanni, T., Sentana, L., Roberts, C., Vedaldi, A., Tolan, J., Brandt, J., Couprie, C., Mairal, J., Jégou, H., Labatut, P., and Bojanowski, P. *DINOv3*. arXiv preprint arXiv:2508.10104, 2025. Available: <https://arxiv.org/abs/2508.10104>
- [3] Stevens, S., Wu, J., Thompson, M. J., Campolongo, E. G., Song, C. H., Carlyn, D. E., Dong, L., Dahdul, W. M., Stewart, C., Berger-Wolf, T., Chao, W.-L., and Su, Y. *BioCLIP: A Vision Foundation Model for the Tree of Life*. IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024. Available: <https://arxiv.org/abs/2311.18803>
- [4] Carion, N., Gustafson, L., Hu, Y.-T., Debnath, S., Hu, R., Suris, D., Ryali, C., Alwala, K. V., Khedr, H., Huang, A., Lei, J., Ma, T., Guo, B., Kalla, A., Marks, M., Greer, J., Wang, M., Sun, P., Rädle, R., Afouras, T., Mavroudi, E., Xu, K., Wu, T.-H., Zhou, Y., Momeni, L., Hazra, R., Ding, S., Vaze, S., Porcher, F., Li, F., Li, S., Kamath, A., Cheng, H. K., Dollár, P., Ravi, N., Saenko, K., Zhang, P., and Feichtenhofer, C. *SAM 3: Segment Anything with Concepts*. arXiv preprint arXiv:2511.16719, 2025. Available: <https://arxiv.org/abs/2511.16719>
- [5] Imageomics Institute. *MMLA Mpala: Multi-Modal Labeled Animals at Mpala*. Hugging Face Datasets, 2024. Available: https://huggingface.co/datasets/imageomics/mmla_mpala
- [6] Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., and Sutskever, I. *Learning Transferable Visual Models from Natural Language Supervision*. International Conference on Machine Learning (ICML), 2021. Available: <https://arxiv.org/abs/2103.00020>
- [7] Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., and Houlsby, N. *An Image is Worth 16×16 Words: Transformers for Image Recognition at Scale*. International Conference on Learning Representations (ICLR), 2021. Available: <https://arxiv.org/abs/2010.11929>

- [8] Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., and Zitnick, C. L. *Microsoft COCO: Common Objects in Context*. European Conference on Computer Vision (ECCV), 2014. Available: <https://arxiv.org/abs/1405.0312>
- [9] Jocher, G., Chaurasia, A., and Qiu, J. *Ultralytics YOLO*. 2023. Available: <https://github.com/ultralytics/ultralytics>
- [10] Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., Desmaison, A., Köpf, A., Yang, E., DeVito, Z., Raison, M., Tejani, A., Chilamkurthy, S., Steiner, B., Fang, L., Bai, J., and Chintala, S. *PyTorch: An Imperative Style, High-Performance Deep Learning Library*. Advances in Neural Information Processing Systems (NeurIPS), 2019. Available: <https://arxiv.org/abs/1912.01703>
- [11] Ohio Supercomputer Center. *Ohio Supercomputer Center*. 1987. Available: <https://www.osc.edu/>